

Ivan Nikolaev

AI Engineer

I break a process into steps and put AI agents where they make a step faster and better.
Under the hood, a senior backend engineer in Go and Ruby on Rails, almost 7 years shipping. Moscow, remote.

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1.57 → 0.49 review comments per MR 40 → 104 merged MRs per half-year
2 → 6 tasks per sprint, median

Main work

Context for agents, not prompts

AI in engineering, day job, 2026

Step: writing code and review. The agent got the team's context: 3× fewer review comments per MR and 2.5× more merged MRs per half-year.

1.57 → 0.49 review comments per MR; 71% → 83% MRs with zero comments

A RAG bot over the wiki, in the team messenger

Day job, in production, 2026

Step: finding out “how does this work here”. Instead of DMs, a bot answers from the wiki for 0.5–1.2 ₺, and quality is measured by evals, not by eye.

13 / 15 → 16 / 17 golden-set eval; 7K, answers silently degraded → 24K plus a recap on overflow context in long threads, tokens

“Review 540”: performance reviews out of Excel

Side project, B2B SaaS, 16 days

Step: scoring performance reviews. HR no longer runs 3,280 rows through Excel formulas, scoring is checked against a reference in tests, and anonymity rests on HMAC.

hand-made Excel formulas → Excel reference locked in a regression test scoring; person id next to the score → HMAC, unrecoverable after retention answer-to-respondent link

“Svod”: documents where styling lives in the template

Side project, closed beta, svod.webtm.ru

Step: formatting documents. Styling is set once in the template, and an agent builds that template from a Word contract on its own via MCP.

in every paragraph → in one template formatting; 214 → fixed in waves audit findings (19 dimensions)

Open source: provenance and database-seed-cli

My own libraries, Ruby and Go, 2026

Step: auditing and test data in new services. Auditing comes from a gem instead of being rewritten, and consistent dev data is generated without a prod dump.

rewritten in every service → gem "provenance" auditing in a Rails app; prod dump or by hand → ~82K rows, zero manual edits dev data

Other work

- **An MCP server for the tracker and wiki.** agent can't see tasks → 22 tools
- **Agent-driven defect analysis.** by hand, quarterly → ~\$0.17 per 24 tasks
- **API descriptions across 10 services.** 0 → 1 100+ parameter descriptions
- **Hermes, a home agent.** 10+ home service web UIs → one Telegram chat, 20 MCP servers
- **Flaky tests show up on the MR.** re-run the pipeline and forget → 15 flaky specs caught on 75 MRs
- **CI tests 2.6× faster.** ~12.5 min → ~4.5 min tests in the pipeline
- **Document-first API tests.** schema from tests → tests from schema
- **Real errors visible in Sentry again.** errors buried under zombie jobs → you see what needs fixing
- **Delivery report.** manual Excel → Grafana
- **Read replicas without surprises.** 805 → 0 violations over 17,207 tests
- **Redis over TLS in 10 services.** token blacklist silently broken → 10 services on TLS, blacklist works
- **Splitting a shared bucket.** 1 bucket, 78 GB → one bucket per service
- **EDUG, a learning bot.** an idea → a working bot and 2nd place

Experience

Nov 2023 – now	Senior Backend Engineer, Lead Software Engineer, ПСБ цифровая лаборатория An education platform, ~30 services in Ruby on Rails and Go, Kafka, PostgreSQL with replicas, k8s. 427 resolved tasks, 346 MRs, third by commits of 59 authors in the main service. Code review, buddy for new hires.
Aug 2020 – Nov 2023	Full-stack Developer, ТренингСпэйс Client projects; a year in, mentor on the largest one: estimates and requirements together with the PM. Later a product team and backend mentoring for the mobile team.
Feb 2022 – Jun 2023	Lecturer, alongside the main job, гбпоу тпск им в.м максимчука Databases, information systems, operating systems. Supervised graduation projects of 20 students, coached students for championships and hackathons.
Oct 2022 – Dec 2022	Instructor, Альтшу Университет Git, programming basics, database design.
Dec 2019 – Aug 2020	Intern developer → Technician, ФГАНУ ЦИТиС Intern developer, then technician. My first Ruby on Rails service, a support desk with server-side rendering.

Skills

Core	Go, Ruby on Rails, PostgreSQL, Kafka, Redis, Sidekiq, REST, GraphQL, OpenAPI
Infra	Docker, Kubernetes, GitLab CI, MinIO, OpenSearch, Grafana, Sentry
Frontend when needed	TypeScript, React
Team	code review, buddy for new hires, backend mentoring, supervised graduation projects of 20 students, WorldSkills coaching
Languages	Russian native, English B1: docs and written communication
AI at work	LLM, AI agents, Claude Code, MCP, tool calling, RAG, evals, custom skills

Championships

WorldSkills Russia, “IT Software Solutions for Business”, 2017–2019. 1 place, Moscow championship; 5 place, national final; 4 place, DigitalSkills. As an expert: compatriot expert at WorldSkills: coached and accompanied 4 competitors; twice invited industry expert at Moscow championships; 50+ demonstration exams as expert and chief expert.

Games

Papenture (Unreal Engine, Unreal MegaJam 2022, a team of 6); **AV** (GameMaker, Major Jam 3, theme “Retro”); **Bull Hellsing** (GameMaker, in development); **Tagman** (Action platformer, in development). itch.io/zander-games

Education

2021–2025	Национальный исследовательский университет «МЭИ» Institute of Radio Engineering and Electronics
2016–2021	ESEDU Computer Science and Engineering
2019	Courses C# 6.0, Microsoft M10262A

Full version with case studies and screenshots: nikolaevyiu.ru. Habr, August 2026, 6.3K reads: habr.com/ru/articles/1075224/